

Imagining the Far East: Exploring Perceived Biases in AI-Generated Images of East Asian Women

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Abstract

Image-generating AI, which allows users to create images from text, is increasingly used to produce visual content. Despite its advancements, cultural biases in AI-generated images have raised significant concerns. While much research has focused on issues within Western contexts, our study examines the perceived biases regarding the portrayal of East Asian women. In this exploratory study, we invited East Asian users to audit three popular models (DALL-E, Midjourney, Stable Diffusion) and identified 18 specific perceived biases, categorized into four patterns: Westernization, overuse or misuse of cultural symbols, sexualization & feminization, and racial stereotypes. This work highlights the potential challenges posed by AI models in portraying Eastern individuals.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**;
• **Social and professional topics** → **Cultural characteristics**;
Race and ethnicity; **Women**.

Keywords

Generative AI, AI Bias, Algorithm Auditing, Algorithm Ethics, Intelligent Communication

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1 Introduction

With the rise of generative artificial intelligence models (Generative AI), content creation is experiencing a revolutionary shift. Among the various forms of generative AI, image-generating AI is one that has matured earlier. In addition to its rapid image generation, image-generating AI has shown strong capabilities in comprehension, visual synthesis, style transfer, and the enhancement of human creativity [50]. Today, image-generating AI has found fertile applications in industries such as art, news media, film production, gaming, and architecture. According to a report by Everypixel Journal in 2023 [19], AI generated 15 billion images within just 18 months, which is equivalent to the number of photos taken by human photographers over the past 150 years. However, image-generating AI also exhibits concerning characteristics. For example, Ghosh and Caliskan [14] found that when requested to draw a “person”, AI was more likely to draw a Western white male. Cheong et al. [9] and Sun et al. [43] discovered that AI models tend to reinforce both gender and racial stereotypes when depicting professions. Such biases have also been reported by media outlets such as The Washington Post [38] and Bloomberg [6], sparking public attention.

However, while previous studies and discussions have laid a solid foundation in bias research, they have predominantly focused on Western contexts (e.g., biases against African Americans) and recruited Western participants. This resonates with previous reflections within the HCI community, which noted that 73% of study findings published at the leading HCI conference, ACM CHI, are based on Western samples, a demographic that represents less than 12% of the world’s population [25]. Thus, in this work, we seek to contribute an Eastern perspective by focusing on a specific theme: the portrayal of East Asian women in AI, a topic that is both classic in research (see Section 2.1) and a real-world concern. For example, in recent years, Western brands such as Dior, Zara, and Dolce & Gabbana have faced severe backlash and commercial value damage due to the inappropriate portrayal of East Asian women in visual media (e.g., Figure 1). Although these brands sometimes explain that the portrayals are not driven by malicious bias, but rather by

a pursuit of art or an attempt to highlight the uniqueness of East Asian women (e.g., [3]), many East Asians indeed feel discomfort or offense due to complex historical, cultural, and political reasons. In the era of image-generating AI, we speculate that similar issues may also be triggered or exacerbated by AI-produced images.



Figure 1: Examples of backlash against Western brands due to their portrayal of East Asian women: (A) Dior faced accusations of racism due to pulled eye advertisements and the uglification of East Asian women [3, 4], (B) A German advertisement showing an East Asian woman getting aroused after smelling dirty laundry worn by white men fuelled anger in Asia [44], (C) Victoria's Secret was repeatedly criticized for cultural appropriation [18].

However, it must be acknowledged that in such incidents, it is challenging to determine whether a controversial portrayal from the West results from intentional bias or merely reflects a lack of understanding of East Asian culture and differing aesthetics. Therefore, rather than distinguishing intentional biases from the producers' side, this work focuses more on the user side and specifically explores the **perceived biases** that East Asian users might report, thus providing insights for the development and refinement of related models and tools. Driven by this motivation, we selected three mainstream image-generating AI models (i.e., DALL-E, Midjourney, Stable Diffusion) and invited East Asian users to perform algorithm audits focused on the theme of creating images of attractive East Asian women. As a result, we identified a total of 18 perceived biases categorized into four patterns, including Westernization, overuse or misuse of cultural symbols, sexualization & feminization, and classic racial stereotypes. We did not examine images of males in this work, as their representation is shaped by distinct interpretive frameworks and origins, and the issue of women's images is currently more prominent in public discussion. For example, historically, East Asian men have been depicted as loyal warriors, chivalrous swordsmen, and kung fu masters, but also as small and weak [1, 29]. A future analysis of how AI generates images of males would be a valuable addition to the field.

Positionality statement: The authors of this work were all born and raised in East Asia. This personal connection to the subject matter has motivated us to critically examine how AI-generated images may perpetuate or challenge existing biases and stereotypes of East Asian women. We recognize that our perspectives are influenced by our upbringing and cultural context, and we strive to approach our research with awareness and sensitivity to these influences. Our goal is to contribute to a more nuanced understanding of AI's impact on representation within our cultural context.

2 Background and Related Work

2.1 East Asian Women in the Eyes of the West

“The West” and “The East” are often used in pairs. Although they may seem to describe physical locations, the division between the East and the West is more about cultural perceptions [33]. The term “West” is primarily based on Western European culture and extends to include regions such as North America and Australia. In contrast, “East” refers to regions not dominated by Western culture, with Asia being a representative example. This vast region can be further divided into distinct cultural groups, such as East Asia, South Asia, and the Middle East. Among them, East Asia (also called Far East), centered around China, Japan, and South Korea, has been deeply influenced by Confucianism, exhibiting cultural orientations distinct from the West [30].

Regarding the appreciation of beauty, while there is a general consensus on aesthetics among humans (e.g., symmetrical and harmonious faces [26]), there are still significant differences in the specific standards of attractiveness for women between the East and the West. For instance, traditional folk stories in East Asia often emphasize facial features over body shape when describing female beauty, with a preference for rounder faces and fairer skin [8]. Psychological research has indicated that East Asians have a lower sensitivity to sexual maturity when evaluating physical attractiveness compared to Western people [11]. Cultural and historical research has revealed that the portrayal of East Asian women has evolved alongside shifting power dynamics between the East and West [42, 51]. In the early 14th century, the West saw China, the giant empire in the East, as a utopia, and greatly admired Chinese aesthetics. However, after 1750, with the rise of Western power, the image of East Asia plummeted [51]. Stereotypes such as “piggy eyes” and “slanted eyes”, along with perceptions of Eastern women as exotic, quiet, passive, and controllable, later permeated Western films, comics, and the fashion industry [1, 32, 47]. However, traditional research on the imagery of East Asian women has predominantly been conducted within the domains of literature, film, history, and political science. Therefore, this study seeks to re-examine this classic subject within the contemporary context of generative AI.

2.2 AI Biases

With the advent of AI, the imperative to develop trustworthy and responsible AI systems and to minimize AI-related harm has become increasingly pressing [10, 22, 23, 41]. AI bias, in particular, has emerged as a significant concern [45]. To date, while various technologies have been developed to identify and mitigate bias [40], those that are more easily addressed often pertain to the more objective *technical biases*. In contrast, *social biases*, which are relevant to human concerns, are more complex, controversial, and challenging to tackle [45]. For example, researchers have detected price discrimination targeting specific regions and social classes on platforms such as Google, Bing, and Amazon [34, 35]. Besides, a set of algorithmic auditing endeavors have shown that women face varying degrees of disadvantages in algorithmic environments [15], such as being assigned less diverse and more stereotypical labels [36, 37],

being less likely than men to be recommended for high-paying positions and roles in the STEM fields [7, 24], and being overclassified as high risk in criminal risk assessments [16].

With the rise of image-generating AI, biases have also been detected in visual forms. For example, Ghosh and Caliskan [14] found that Stable Diffusion was more likely to generate images of white males when prompted with the term “person”. Cheong et al. [9] discovered that DALL-E defaulted to portraying white individuals in occupational images, with only a few professions, such as rapper, featuring more Black individuals. Sun et al. [43] found that AI exhibits not only *representation bias* (e.g., depicting “housekeeping” as female and “scientist” as male) but also *presentation bias*, such as portraying women more often as happy, smiling, and in submissive positions. Mack et al. [31] found that AI often oversimplifies the depiction of people with disabilities and displays negative stereotypes (e.g., being in a wheelchair or appearing helpless). Karpouzis [21] drew on Plato’s *Allegory of the Cave* and commented that cultural biases in AI could exacerbate the echo chamber effect, where mainstream culture is amplified while minority voices are weakened or silenced. Hence, there is a pressing need for more empirical research to deepen our understanding of image-generating AI.

3 Exploring Perceived Biases in the Portrayal of East Asian Women

In this section, we report an exploratory study that examines whether and how users perceive biases regarding East Asian women drawn by image-generating AI models.

3.1 Methodology

3.1.1 Algorithm audit. Algorithm audit is an empirical research method used to investigate potential risks in algorithmic systems, particularly the manifestation of social problems [2]. A crowd-sourced audit is a user-driven method [12] that recruits real-world users to audit algorithms and gather feedback to uncover algorithmic issues. This method is particularly suitable for exploratory studies. Therefore, we finally chose to invite East Asian users to audit three mainstream image-generating AI models: (i) DALL-E 3 model by OpenAI (USA), (ii) Midjourney v6 model by Midjourney Lab (USA), and (iii) Stable Diffusion 3 model by Stability AI (UK). These models represent the most widely used and highest-output text-to-image AI models [19].

3.1.2 Participants. We recruited participants by posting advertisements on social media and snowballing. The recruitment materials specified the following requirements: (i) Participants should come from East Asian countries (China, Japan, South Korea), predominantly reside within East Asia, and self-identify as East Asian. (ii) Participants should have knowledge and some experience with image-generating AI. (iii) Participants should be proficient in English to operate English interfaces. A total of 36 qualified participants were recruited (25 females, 11 males), including 20 Chinese, 7 Japanese, and 9 Koreans. Their ages ranged from 18 to 50 ($M = 23$, $SD = 6.3$). Since the study required the participants to navigate English interfaces and write prompts in English, all participants had college education (Bachelor’s Degree: 88.89%, Master’s Degree: 11.11%). Their educational backgrounds were diverse, encompassing fields such as social sciences, engineering, medicine, and management.

The study was conducted offline in China with each participant individually. All participants received a compensation of \$8.

3.1.3 Study Procedure. To start with, we introduced the study’s background and provided a tutorial on DALL-E, Midjourney, and Stable Diffusion. After the participants became familiar with these tools, they were tasked with “drawing an attractive East Asian woman” by exploring and using each tool to perform the task. We used the term, “attractive” (i.e., a person’s physical features are considered aesthetically pleasing or beautiful [27]), following the established research tradition in psychology that has extensively studied physical attractiveness since the 20th century (e.g., [5, 11, 26, 46]). Compared to words such as “beautiful” and “good-looking”, this term not only encompasses multiple dimensions, such as facial features, body proportions, and skin condition, but also raises fewer ethical concerns, as it is less likely to be perceived as judgmental or discriminatory. For each generated image, they were asked to record the image, the tool used, and the prompt in a spreadsheet. To help the participants get started, we provided some pre-prepared keywords, including (i) adjectives such as “attractive”, “beautiful”, and “pretty”¹; (ii) adverbs describing intensity, such as “very” and “highly”; (iii) nouns relating to women, such as “female” and “lady”; (iv) nouns indicating regions, such as “East Asia”, “China”, “Japan”, and “Korea”. These keywords were merely provided for inspiration, and participants were encouraged to write their own prompts as they wished. We also provided participants with an English dictionary and a thesaurus tool to help them write their own prompts.

Next, we conducted interviews with each participant, focusing on questions such as: “What should be the characteristics of an attractive East Asian woman?”, “Could you provide a photo representing your ideal woman image?”. Besides, we asked the participants to go through all the images they generated one by one, explaining whether they perceived any biases, and why. Lastly, the participants rated the three models in terms of their overall satisfaction on a scale from 0 to 100. The interviews were audio recorded with the participants’ consent.

3.2 Analyses and Findings

We collected 675 AI generated images from the study (i.e., 18.75 images per participant on average) and 36 images of ideal East Asian women provided by our participants. We also obtained an abundance of qualitative data elucidating the participants’ opinions and explanations. Below, we report our data analyses and major findings. Note that when quoting participants, we use the format P1-P36 to refer to their IDs, with subscripts indicating their nationality (CN: China, JP: Japan, KR: Korea) and the tool they commented on in the quote (DE: DALL-E; MJ: Midjourney; SD: Stable Diffusion). For example, P1_{CN,DE} refers to Participant 1, who is Chinese and commented on DALL-E.

3.2.1 Ideal images. We analyzed the participants’ qualitative descriptions of their ideal East Asian woman images using natural language processing techniques. The most frequently mentioned

¹Mainstream image-generating AI platforms generally have ethical checks in place. Our pre-study test showed that if users input insulting or offensive terms, such as “ugly”, the platforms either refuse to create the image or convey negativity by only adjusting clothing or distorting facial features. Therefore, in this study, prompts used only unidirectional adjectives to ensure a focus on creating attractive women.

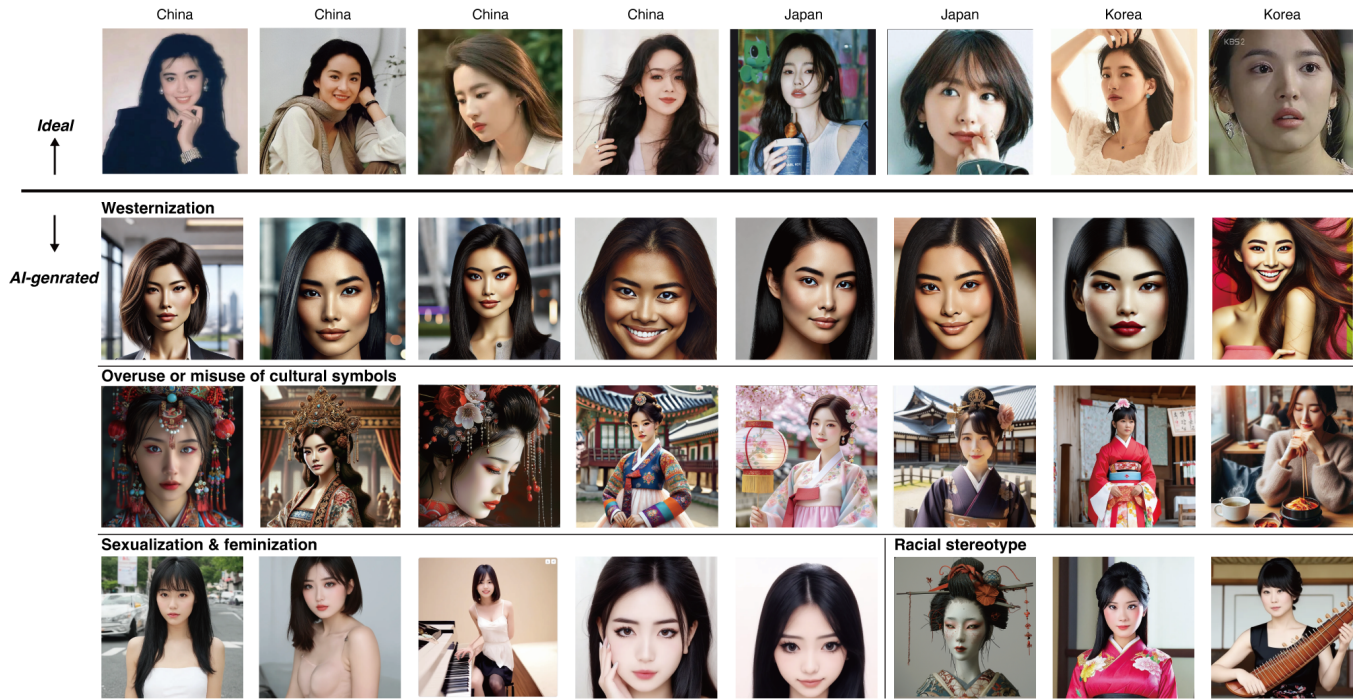


Figure 2: Examples of ideal images of East Asian women provided by our participants, and examples of AI-generated images.

nouns associated with ideal women were *eyes* ($N = 16$), *facial features/face* (16), *temperament* (9), *style* (6), and *hair* (6). The most mentioned adjectives were *overall (feeling)* (8), *soft/gentle* (8), *fair (skin)* (7), *well-proportioned/harmonious* (6), *natural* (4), and *healthy* (4). Overall, the participants' ideal images of East Asian women were highly consistent with the indigenous beauty standards identified in previous literature [8, 13, 17, 20], which emphasize facial features (particularly the eyes) over body shape, and prefer roundness over sharpness. For example, our participants described their ideal image of a woman as “*having bright eyes*” ($P5_{CN}$); the eyes “*don't need to be very large*” ($P34_{KR}$) and “*should be luminous*” ($P3_{CN}$). There was also a pursuit of fair skin (e.g., $P21_{JP}$: “*a bit whiter than yellow*”, $P27_{CN}$: “*skin should be fair but not as white as Westerners*”), as well as a pursuit of harmonious and round facial features (e.g., $P13_{CN}$: “*no sharp angles*”, $P32_{KR}$: “*overall balance is more important than the beauty of each individual facial feature.*”) Echoing previous finding [11], our participants were not sensitive to features associated with sexual maturity. None of them mentioned keywords such as “*sexy*”. Most participants preferred a softer and elegant temperament (e.g., $P28_{JP}$: “*not too fierce, more gentle*”, $P19_{CN}$: “*elegant and poised, appearing very dignified and graceful*”, $P35_{KR}$: “*I like women who are polite and cultured*”). Besides, the participants generally favored natural beauty, and five of them had intentionally emphasized the avoidance of heavy makeup (e.g., $P4_{CN}$: “*neat and natural, absolutely no heavy makeup*”, $P24_{JP}$: “*looking natural, without complicated makeup*”).

However, many AI-generated images deviated from their ideals. Only 43.7% of the generated images were reported as satisfactory by

our participants (49.6% of DALL-E images, 52.7% of Midjourney images, and 14.4% of Stable Diffusion images). Participants frequently had to fine-tune the AI through multiple rounds to obtain a relatively satisfactory image. None of them received a satisfactory image right from the start.

3.2.2 Taxonomy of Perceived Biases. Next, we analyzed whether participants reported any perceived biases in AI-generated images with two research questions in mind: (i) *where* the biases occurred, and (ii) *how* the participants interpreted the biases. We adopted the open coding method. Two coders first independently performed their coding. For (i), we generated codes based on the specific features that the participants identified as biased. For example, some participants reported perceived biases regarding *lips* and *skin*. For (ii), we coded participants' explanations and justifications of the biases, revealing why they perceived a certain feature as biased. For example, when reporting their unease about thick lips and heavy makeup, the participants often provided interpretations such as “*It suddenly reminds me of those Asian characters seen in American dramas*” and “*It looks more like a Western style*”. Thus, we grouped these codes into a higher-level category called *Westernization* to indicate their pattern of bias. Next, the two coders met to compare their codes, unify the naming of similar codes, and discuss any inconsistencies. As shown in Table 1, 18 reported biases were identified, categorized into four patterns. Examples of these patterns are shown in Figure 2.

Westernization. Westernization refers to the portrayal of East Asian women in ways that cater to Western physical appearance, styles, and values (see Figure 2, the second row). Within this pattern, the most mentioned bias is the use of Western-style makeup.

Bias patterns	Issues	Models Description
Westernization (144)	western-styled makeup (27)	  thick eyelashes and eyeshadow, broad eyebrow shapes, and heavy blush
	bony/sharp face (24)	  Salient bone structure, such as high cheekbones and a pointed chin
	eyes (22)	  Deep eyes, long eye fissures, double eyelids, short intercanthal distances
	nose (21)	  A high nose, narrow nasal widths
	lips (21)	  Full lips (or using lipsticks to emphasize full lips)
	skin tone (15)	  Skin color leans towards the tanned wheat and chocolate shades
	big smile (14)	  A broad smile showing teeth, with smile muscles engaged
Sexualization & feminization (64)	revealing clothing (15)	  Clothing that reveals breasts or buttocks, such as low-cut tops, deep V-necks, and mini skirts
	porcelain skin (12)	  Highly smooth and flawless skin, lacking wrinkles and texture
	youth (10)	  Drawing young women by default; difficult to portray beautiful middle-aged or elderly women
	dull eyes (9)	  The gaze is dull, lacking confidence, avoiding direct eye contact
	harsh gaze (9)	  The gaze is fierce, piercing, and dangerous
	slenderness (5)	  Drawing a slender figure by default; difficult to portray plump beauties
Overuse/misuse of cultural symbols (63)	long hair (4)	  Drawing long hair that extends beyond the shoulders by default
	ancient/exotic symbols (25)	  Placing women in exotic backgrounds or dressing them in traditional clothing, jewelry, and patterns
	mix of cultural elements (22)	  Inability to distinguish between the differences within East Asian countries, leading to mixed use of elements
Racial stereotype (13)	wrong semantics (16)	  Not in line with the real environment and living habits of East Asia
	slanted eyes (13)	  Eyes that are narrow in width and angled upward at the outer corners (or makeup that emphasizes this trait).

Table 1: A taxonomy of the perceived biases identified in our study. The “Models” column shows the relative distribution of these biases across the three models.  : DALL-E,  : Midjourney,  : Stable Diffusion.

In these images, although the women have East Asian faces, the makeup is applied to align their appearance more closely with Western preferences for bone structure and sexual maturity [11]. This practice deviates from the East Asian aesthetic preference for minimal or natural makeup (as reported in Section 3.2.1). For example, $P22_{KR,DE}$ said, “The eye makeup is very deep and heavy; I don’t think ordinary Koreans wear makeup like this.” $P11_{CN,DE}$ complained that “I just feel that her eyebrows are too thick.” She then tried to fine-tune the prompts and found that “there wasn’t any substantial improvement.” $P21_{JP,MJ}$ also commented that “the makeup it produces feels very foreign for me.” Apart from makeup, our participants also reported instances where AI-generated images drew East Asian women with Western physical traits, such as eyes, noses, lips, and bony/sharp faces. For example, $P22_{KR,MJ}$ commented, “The facial features are too Western; the eyes seem too large.” $P10_{CN,DE}$ noted that the image she received “looks like an American-born Asian, and the cheekbones remind me of Kim Kardashian.” Such blending of Western features with East Asian facial structures often results in what is termed a pan-Asian face, which typically suggests a White ancestor in the person’s family tree [49]. In Asia, this phenomenon is more common in countries with a history of significant Western colonization and racial mixing, such as those in South and Southeast Asia [28, 39]. Therefore, we also observed complaints such as, “Although the girl is also beautiful, her appearance clearly has a Southeast Asian style” ($P3_{CN,SD}$) and “The face shape looks like that of a foreigner, specifically a Thai person” ($P25_{JP,DE}$). $P4_{CN,DE}$ said, “Look at the skin color, it’s not East Asian; it’s very dark, like a mixed-race with Black and White ancestry.” Additionally, several participants reported perceived bias related

to the depiction of a big smile. For example, $P3_{CN,SD}$ noted that showing a lot of teeth is not a common or ideal expression for East Asian women, but the model “always produces a very exaggerated big smile.” $P6_{CN,DE}$ attributed such portrayals to Western culture, which emphasizes confidence: “The smiling muscles are too intense and exaggerated, resembling that of a very confident and sunny ABC (American-born Chinese).”

Sexualization & feminization. In our study, many participants reported uneasiness of seeing East Asian women portrayed with revealing clothing. For example, $P12_{CN,SD}$ was surprised by the output image of Stable Diffusion: “I didn’t expect it to be a girl like this, who gives off the impression of a woman of the street, and her clothing is also quite revealing.” When an image of a woman in a deep V-neck outfit was generated, $P22_{KR,DE}$ said, “We’re not fond of that style; its neckline is a bit too low.” $P25_{JP,DE}$ commented that they usually “wouldn’t wear a V-neck that plunges so low, and a tight skirt; it doesn’t seem to represent East Asia well.” Participants also noted several features that emphasize women’s passivity and obedience, such as porcelain skin, dull eyes, and portrayals of them as consistently thin, young, and with long hair. For example, $P5_{CN,MJ}$ attempted to draw a middle-aged Chinese woman using the three models but was frustrated to find that “these models seem not to understand how women can maintain their beauty as they age. They either ignore my prompts and insist on drawing young girls, or they just awkwardly add a few wrinkles.” $P1_{CN,DE}$ wanted to draw a woman with a plump figure, which used to be a sign of wealth and health in East Asia. She tried keywords such as “plump”, “fat”, “chubby”, as well as more specific weight descriptors like “60kg”, “80kg”. However, the model continued to produce images of slim

women. $P9_{CN,SD}$ found that in some images, “the eyes seem lifeless, and to put it bluntly, it’s as if the eyes suggest a lack of intelligence.” $P6_{CN,SD}$ said, “the gaze is very rigid, which fits the discriminatory stereotype of East Asians by Westerners - thinking we are not wise enough, more docile, and obedient.” Moreover, while participants generally prefer fair skin, they disliked the doll-like, artificial paleness (e.g., $P13_{CN,SD}$: “It might perceive East Asian women as somewhat petite and delicate, or as having a kind of pitiable charm”). Several participants thought depicting East Asian women uniformly with long hair is biased ($P4_{CN,DE}$: “All have straight, black, long hair... If everyone is generated to be like this, I think it is a bias.”) The aforementioned features closely resemble stereotypical archetypes once identified in Western narratives, such as the China Doll and the Geisha Girl [48]. Besides, we also observed a few cases related to another sexualized archetype, namely the Dragon Lady, a mysterious and dangerous woman who allures men. Our participants primarily perceived this pattern through the women’s harsh gaze (e.g., $P8_{CN,DE}$: “It feels a bit too severe, I mean her temperament”; $P20_{CN,MJ}$: “The gaze is quite fierce, not nice.”)

Overuse or misuse of cultural symbols. The third reported pattern is the overuse or misuse of cultural symbols (Figure 2, third row). This includes the awkward and deliberate integration of ancient symbols (e.g., traditional headdresses, earrings, clothing patterns, umbrellas, lanterns), as well as positioning East Asian women in front of traditional architecture like temples and gardens. For instance, $P1_{CN,MJ}$ noted that Midjourney “generates images of ancient clothing, which feels like a stereotype. When I asked for modern clothes, it still provided me with ancient costumes.” $P8_{CN,DE}$ questioned: “Why do they have to wear that kind of traditional clothing? I find it quite strange.” $P26_{JP,MJ}$ generated an image of a Japanese mother playing with her children but was depicted in a kimono, which she found odd: “It is weird, as mothers generally do not wear kimonos to parks.” $P27_{KR,DE}$ drew a Korean girl shopping, but the generated background was a vintage shop selling traditional Korean dress: “It is not a typical place we would go to.” The second perceived bias is the confounding use of cultural elements, which fails to distinguish between the cultural differences within East Asia. For instance, $P21_{JP,DE}$ reported a case where she attempted to draw a Japanese woman, but the output “looks very much like my perception of women in Chinese period dramas, but I wrote that she should be Japanese.” $P25_{JP,MJ}$, intending to draw a Japanese individual as well, said, “Seeing the costume, I felt momentarily bewildered. This should be a Hanfu dress from China, what’s the situation?” $P22_{KR,MJ}$ complained that “South Korea never had umbrellas like this; these are entirely Japanese.” The third bias is the wrong semantics, that is, the application of symbols deviates from the actual situation. For example, $P23_{KR,DE}$ generated an image depicting a woman eating, but observed, “When consuming this type of meal in South Korea, a spoon is typically used, yet the image showed chopsticks.” $P18_{CN,DE}$ requested an AI-generated image of a beautiful woman from northern China but found that the background elements are all from the scenery of southern provinces: “There’s no such terrain in the north.” The way AI mishandles East Asian cultural symbols echoes what Ziff and Rao [52] termed cultural appropriation: the incorporation of East Asian cultural symbols by AI is not based on genuine understanding but is primarily used to create an exotic atmosphere. These symbols are often assembled in an arbitrary

and decontextualized manner, resulting in portrayals that seem unnatural and disjointed.

Racial stereotype. Several participants spotted the classic racial stereotype of slanted eyes. $P25_{JP,MJ}$ found that “the eyes seem squinty, a typical slanted eyes.” $P8_{CN,DE}$ commented, “Monolid eyes, upturned eye shapes; it may not rise to the level of racial discrimination, but it could indeed be a stereotype.” In East Asia, slanted eyes are not merely a biological trait; they are also associated with the collective trauma stemming from East Asia’s history of being discriminated against and exploited by the West. However, this pattern’s occurrence in AI-generated images is relatively low compared to past patterns. We speculate this is due to the strict ethical reviews on many AI platforms. Yet, our participants were notably sensitive to and immediately noticed the shape of slanted eyes in images once it emerged. Although some participants acknowledged that certain East Asians do have such eye shapes, they hoped that AI would not emphasize these eyes or would draw more aesthetically pleasing eyes (e.g., $P22_{KR,DE}$: “I feel uncomfortable seeing this; or, let’s say, at least I don’t find them beautiful. I think it overly emphasizes the upward slant of the eyes of East Asians.” $P26_{JP,SD}$: “In my life, most people do not have this eye shape.”)

3.2.3 Observations. As shown in Table 1, the distribution of identified biases varied across models. Of the 18 biases, 9 (50.00%) were most frequently observed in DALL-E, 6 (33.33%) in Stable Diffusion, 2 (11.11%) in Midjourney, and 1 (5.56%) was equally observed in DALL-E and Stable Diffusion. The three models also exhibited different distributions across the four bias patterns. DALL-E was most frequently associated with *Westernization* biases. Midjourney was most frequently reported for biases related to the *overuse/misuse of cultural symbols*. For biases related to *sexualization & feminization* and *racial stereotypes*, Stable Diffusion was the most frequently reported model. Overall, participants expressed the highest satisfaction with Midjourney (mean satisfaction score = 72.2 out of 100), followed by DALL-E (mean satisfaction score = 62.8). Despite having fewer dominating biases than DALL-E, Stable Diffusion often portrayed East Asian women in ways that were overly revealing, ingratiating, and with slanted eyes, which led to strong disapproval from participants, resulting in the lowest satisfaction rating (mean satisfaction score = 52.3).

4 Conclusion

In this work, we explore the portrayal of East Asian women in three image-generating AI models (DALL-E, Midjourney, Stable Diffusion) through a user-centered auditing study. As a result, we identified a range of perceived biases in AI-generated images of East Asian women. While racial stereotypes were relatively infrequent, biases manifested in various other ways, causing the AI-generated images to diverge from the ideals and expectations of East Asian users. Besides, the manifestation of these issues varied across different models: DALL-E exhibited the most notable Westernization of East Asian women, Stable Diffusion showed the most significant sexualization, and Midjourney had considerable issues with the misuse of cultural symbols. Based on these findings, we advocate for more respectful and inclusive representation in future models.

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References

- [1] Sameena Azhar, Antonia RG Alvarez, Anne SJ Farina, and Susan Klumpner. 2021. “You’re so exotic looking”: An intersectional analysis of Asian American and Pacific Islander stereotypes. *Affilia* 36, 3 (2021), 282–301.
- [2] Jack Bandy. 2021. Problematic machine behavior: A systematic literature review of algorithm audits. In *Proceedings of the ACM on Human-Computer Interaction*. ACM, 1–34.
- [3] BBC. 2021. Photographer sorry for ‘small eyes’ Dior picture. <https://www.bbc.com/news/world-asia-china-59397737>. Last accessed: Jan 5, 2025.
- [4] BBC. 2023. Dior accused of racism over ‘pulled eye’ advertisement. <https://www.bbc.com/news/world-asia-china-65229901>. Last accessed: Sept 1, 2024.
- [5] Ellen Berscheid, Karen Dion, Elaine Walster, and G William Walster. 1971. Physical attractiveness and dating choice: A test of the matching hypothesis. *Journal of experimental social psychology* 7, 2 (1971), 173–189.
- [6] Bloomberg. 2023. Humans are biased. Generative AI is even worse. <https://www.bloomberg.com/graphics/2023-generative-ai-bias/>. Last accessed: Sept 1, 2024.
- [7] Miranda Bogen and Aaron Rieke. 2018. Help wanted: An examination of hiring algorithms, equity, and bias. <https://apo.org.au/node/210071>. Last accessed: Sept 1, 2024.
- [8] Leah Cashin. 2024. The Evolution of Chinese Beauty. *Articulate* 29, 1 (2024), 6.
- [9] Marc Cheong, Ehsan Abedin, Marinus Ferreira, Ritsaart Reimann, Shalom Chalson, Pamela Robinson, Joanne Byrne, Leah Ruppanner, Mark Alfano, and Colin Klein. 2023. Investigating gender and racial biases in DALL-E Mini Images. *ACM Journal on Responsible Computing* 1, 2 (2023), 1–20.
- [10] Rachel Courtland. 2018. The bias detectives. *Nature* 558, 7710 (2018), 357–360.
- [11] Michael R Cunningham, Alan R Roberts, Anita P Barbee, Perri B Druen, and Cheng-Huan Wu. 1995. “Their ideas of beauty are, on the whole, the same as ours”: Consistency and variability in the cross-cultural perception of female physical attractiveness. *Journal of Personality and Social Psychology* 68, 2 (1995), 261.
- [12] Alicia DeVos, Aditi Dhabalia, Hong Shen, Kenneth Holstein, and Motahhare Eslami. 2022. Toward User-Driven Algorithm Auditing: Investigating users’ strategies for uncovering harmful algorithmic behavior. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, 1–19.
- [13] Dimitre Dimitrov and George Kroumpouzos. 2023. Beauty perception: a historical and contemporary review. *Clinics in Dermatology* 41, 1 (2023), 33–40.
- [14] Sourojit Ghosh and Aylin Caliskan. 2023. ‘Person’ == Light-skinned, Western Man, and Sexualization of Women of Color: Stereotypes in Stable Diffusion. In *Conference on Empirical Methods in Natural Language Processing*. ACL, 6971–6985.
- [15] Paula Hall and Debbie Ellis. 2023. A systematic review of socio-technical gender bias in AI algorithms. *Online Information Review* 47, 7 (2023), 1264–1279.
- [16] Melissa Hamilton. 2019. The sexist algorithm. *Behavioral sciences & the law* 37, 2 (2019), 145–157.
- [17] Mijeong Han. 2003. Body image dissatisfaction and eating disturbance among Korean college female students: Relationships to media exposure, upward comparison, and perceived reality. *Communication Studies* 54, 1 (2003), 65–78.
- [18] Business Insider. 2017. Every time Victoria’s Secret has been accused of cultural appropriation in its annual fashion show. <https://www.businessinsider.com/victorias-secret-fashion-show-accused-cultural-appropriation-2017-11>. Last accessed: Sept 1, 2024.
- [19] Everypixel Journal. 2023. AI Has Already Created As Many Images As Photographers Have Taken in 150 Years. Statistics for 2023. <https://journal.everypixel.com/ai-image-statistics>. Last accessed: Sept 1, 2024.
- [20] Jaehye Jung and Seung-Hee Lee. 2006. Cross-cultural comparisons of appearance self-schema, body image, self-esteem, and dieting behavior between Korean and US women. *Family and Consumer Sciences Research Journal* 34, 4 (2006), 350–365.
- [21] Kostas Karpouzis. 2024. Plato’s Shadows in the Digital Cave: Controlling Cultural Bias in Generative AI. *Electronics* 13, 8 (2024), 1457.
- [22] Davinder Kaur, Suleyman Uslu, Kaley J Rittichier, and Arjan Durrezi. 2022. Trustworthy artificial intelligence: a review. *ACM computing surveys (CSUR)* 55, 2 (2022), 1–38.
- [23] Celeste Kidd and Abeba Birhane. 2023. How AI can distort human beliefs. *Science* 380, 6651 (2023), 1222–1223.
- [24] Anja Lambrecht and Catherine Tucker. 2019. Algorithmic bias? An empirical study of apparent gender-based discrimination in the display of STEM career ads. *Management Science* 65, 7 (2019), 2966–2981.
- [25] Sebastian Linxen, Christian Sturm, Florian Brühlmann, Vincent Cassau, Klaus Opwis, and Katharina Reinecke. 2021. How weird is CHI? In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, 1–14.
- [26] Anthony C Little, Benedict C Jones, and Lisa M DeBruine. 2011. Facial attractiveness: evolutionary based research. *Philosophical Transactions of the Royal Society B: Biological Sciences* 366, 1571 (2011), 1638–1659.
- [27] Jie Liu. 2020. Personality and Physical Attractiveness. In *Encyclopedia of Personality and Individual Differences*. Springer, 3669–3675.
- [28] Craig A Lockard. 2009. *Southeast Asia in world history*. Oxford.
- [29] Alexander Lu and Y Joel Wong. 2013. Stressful experiences of masculinity among US-born and immigrant Asian American men. *Gender & Society* 27, 3 (2013), 345–371.
- [30] Jackson G Lu, Richard E Nisbett, and Michael W Morris. 2022. The surprising underperformance of East Asians in US law and business schools: The liability of low assertiveness and the ameliorative potential of online classrooms. *Proceedings of the National Academy of Sciences* 119, 13 (2022), e2118244119.
- [31] Kelly Avery Mack, Rida Qadri, Remi Denton, Shaun K Kane, and Cynthia L Bennett. 2024. “They only care to show us the wheelchair”: Disability representation in text-to-image AI models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–23.
- [32] Gina Marchetti. 1994. *Romance and the yellow peril: Race, sex, and discursive strategies in Hollywood fiction*. Univ of California Press.
- [33] William H McNeill. 1997. What we mean by the west. *Orbis* 41, 4 (1997), 513–525.
- [34] Jakub Mikians, László Gyarmati, Vijay Erramilli, and Nikolaos Laoutaris. 2012. Detecting price and search discrimination on the internet. In *Proceedings of the ACM workshop on hot topics in networks*. ACM, 79–84.
- [35] Jakub Mikians, László Gyarmati, Vijay Erramilli, and Nikolaos Laoutaris. 2013. Crowd-assisted search for price discrimination in e-commerce: First results. In *Proceedings of the ACM conference on Emerging networking experiments and technologies*. ACM, 1–6.
- [36] Safiya Umoja Noble. 2018. *Algorithms of oppression: How search engines reinforce racism*. New York University Press.
- [37] Orestis Papakyriakopoulos and Arwa M Mboya. 2023. Beyond algorithmic bias: A socio-computational interrogation of the Google search by image algorithm. *Social Science Computer Review* 41, 4 (2023), 1100–1125.
- [38] The Washington Post. 2023. This is how AI image generators see the world. <https://www.washingtonpost.com/technology/interactive/2023/ai-generated-images-bias-racism-sexism-stereotypes/>. Last accessed: Sept 1, 2024.
- [39] John F Riddick. 2006. *The history of British India: a chronology*. Bloomsbury Publishing USA.
- [40] Drew Roselli, Jeanna Matthews, and Nisha Talagala. 2019. Managing bias in AI. In *Companion proceedings of the world wide web conference*. 539–544.
- [41] Renee Shelby, Shalaleh Rismani, Kathryn Henne, Ajung Moon, Negar Rostamzadeh, Paul Nicholas, N’Mah Yilla-Akbari, Jess Gallegos, Andrew Smart, Emilio Garcia, et al. 2023. Sociotechnical harms of algorithmic systems: Scoping a taxonomy for harm reduction. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 723–741.
- [42] Jonathan D Spence. 1999. *The Chan’s great continent: China in western minds*. WW Norton & Company.
- [43] Luhang Sun, Mian Wei, Yibing Sun, Yoo Ji Suh, Liwei Shen, and Sijia Yang. 2024. Smiling women pitching down: auditing representational and presentational gender biases in image-generative AI. *Journal of Computer-Mediated Communication* 29, 1 (2024), zmad045.
- [44] The Straits Times. 2019. ‘Racist’ German advertisement infuriates women in South Korea. <https://www.straitstimes.com/asia/east-asia/racist-german-advertisement-infuriates-women-in-south-korea>. Last accessed: Sept 1, 2024.
- [45] Inga Ulinicane and Aini Aden. 2023. Power and politics in framing bias in Artificial Intelligence policy. *Review of Policy Research* 40, 5 (2023), 665–687.
- [46] Elaine Walster, Vera Aronson, Darcy Abrahams, and Leon Rottman. 1966. Importance of physical attractiveness in dating behavior. *Journal of personality and social psychology* 4, 5 (1966), 508–516.
- [47] Jun Xing. 1998. *Asian America through the lens: History, representations, and identity*. Vol. 3. Rowman Altamira.
- [48] Traise Yamamoto. 2000. In/visible difference: Asian American women and the politics of spectacle. *Race, Gender & Class* 7, 1 (2000), 43–55.
- [49] Jeaney Yip, Susan Ainsworth, and Miles Tycho Hugh. 2019. Beyond whiteness: Perspectives on the rise of the Pan-Asian beauty ideal. In *Race in the Marketplace: Crossing Critical Boundaries*. Springer, 73–85.
- [50] Eric Zhou and Dokyun Lee. 2024. Generative artificial intelligence, human creativity, and art. *PNAS Nexus* 3, 3 (2024), pgae052.
- [51] Ning Zhou. 2006. *Western images of China*. BEIJING BOOK CO. INC.
- [52] Bruce Ziff and Pratima V Rao. 1997. Introduction to cultural appropriation: A framework for analysis. In *Borrowed power: Essays on cultural appropriation*. Rutgers University Press New Brunswick, NJ.